ANALYSIS OF DATA COMMUNICATION IN WEB APPLICATION
USING WEB DEVELOPMENT TOOLS

Aim

To analyze and understand the process of data communication in a web application using web development tools. The study focuses on how data is exchanged between the client, web server, and database through HTTP/HTTPS requests and responses. It also aims to examine network requests, response data, status codes, headers, and communication performance using browser Developer Tools.

Objectives

- To understand the basic concept of data communication in web applications.
- To study the client-server architecture used in modern web applications.
- To understand how browsers communicate with web servers using HTTP and HTTPS protocols.
- To analyze different HTTP methods such as GET, POST, PUT, and DELETE.
- To examine how data is transferred between the frontend and backend.
- To understand commonly used data formats such as JSON and HTML.
- To analyze HTTP response status codes such as 200, 404, and 500.
- To study request and response headers used during communication.
- To use browser Developer Tools to monitor network activities.
- To identify possible communication errors and performance issues.
- To understand how efficient data communication improves the performance and reliability of a web application.

Problem Statement

Web applications continuously exchange data between the user's browser and the server. When a user opens a webpage, submits a form, logs in, searches for information, or requests data, the browser sends a request to the server. The server processes the request and sends an appropriate response back to the browser.

If the communication between the client and server is not properly designed, several problems can occur, such as slow loading, failed requests, incorrect data, communication errors, large data transfers, and security risks. Therefore, there is a need to analyze the data communication process of a web application and identify how requests and responses are transmitted, processed, and received. Web development tools can be used to monitor this process and determine the performance and correctness of communication.

Proposed Solution

A sample web application is considered for analyzing the communication process. The application consists of a frontend, backend/server, and database/API. The frontend can be developed using HTML, CSS, and JavaScript. When the user performs an action, JavaScript sends a request to the backend server using HTTP/HTTPS. The server processes the request and communicates with the database or API if required. Finally, the server sends the response back to the browser.

Data Communication Flow:

User

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Web Browser / Client

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HTTP/HTTPS Request

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Web Server / Backend

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Database / API

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Server Processes Data

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HTTP/HTTPS Response

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Web Browser

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Result Displayed to User

Web Development Tools Used

a. Browser Developer Tools

Developer Tools available in browsers such as Google Chrome and Microsoft Edge can be used to inspect the communication between the client and server.

b. Network Tab

The Network tab is used to monitor all network requests made by the webpage. It provides information about the request URL, request method, status code, response data, request and response headers, loading time, data transfer size, and type of requested resource.

c. HTML, CSS and JavaScript

HTML is used to create the structure of the webpage, CSS is used for styling, and JavaScript is used to perform dynamic operations and communicate with the server.

d. API

An API (Application Programming Interface) allows the frontend to communicate with backend services and retrieve or send data.

Data Communication Process

Step 1 – User Request

The user performs an action such as clicking a button, submitting a form, or searching for information.

Step 2 – Client Request

The browser generates an HTTP/HTTPS request and sends it to the appropriate server.

Step 3 – Server Processing

The server receives the request, validates the data, performs the required operation, and may communicate with a database.

Step 4 – Server Response

After processing the request, the server sends a response to the browser.

Step 5 – Displaying the Result

The browser receives the response and displays the required information to the user.

HTTP Methods Analyzed

HTTP Method	Purpose
GET	Retrieve data from the server
POST	Send or create new data
PUT	Update existing data
DELETE	Delete data from the server

For example, when a user searches for information, a GET request may be sent to retrieve the required data. When a user submits registration information, a POST request may be used.

HTTP Status Codes

- 200 – OK: Request was successfully completed.
- 201 – Created: New data was successfully created.
- 400 – Bad Request: The request contains invalid data.
- 401 – Unauthorized: Authentication is required.
- 403 – Forbidden: Access to the resource is not permitted.
- 404 – Not Found: Requested resource could not be found.
- 500 – Internal Server Error: An error occurred on the server.

Data Formats

Web applications commonly exchange data in JSON (JavaScript Object Notation) format because it is lightweight and easy for both humans and machines to understand.

```
{
  "name": "Bino",
  "department": "Mechatronics",
  "year": "First Year"
}
```

The browser can send this data to the server or receive similar data from an API.

Analysis Using Network Tool

The browser's Network tab is used to observe the communication process. When the application performs an operation, the corresponding request appears in the Network panel. The following parameters can be analyzed:

- Request method
- Request URL
- Status code
- Response time
- Request payload
- Response data
- Headers
- Transferred data size
- Content type

This helps developers identify slow requests, failed connections, incorrect API calls, and server-side errors.

Advantages of the Analysis

- Helps understand client-server communication.
- Makes it easier to identify network errors.
- Helps improve webpage loading speed.
- Helps developers debug API-related problems.
- Provides information about request and response data.
- Helps identify unnecessary data transfers.
- Improves application performance.
- Helps in designing efficient web applications.
- Supports better understanding of HTTP/HTTPS communication.

Result

The data communication process of the web application was successfully analyzed using web development tools. The interaction between the client, server, API, and database was studied through the browser's Network Developer Tools.

HTTP methods such as GET and POST, response status codes, headers, request data, response data, and network performance were successfully observed. The analysis helped identify how information is transferred from the browser to the server and how the processed result is returned to the user.

Thus, the case study provided a clear understanding of data communication in web applications and demonstrated how web development tools can be used to monitor, debug, and improve the efficiency of client-server communication.